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Microbial community structure in restored riparian soils of the Canadian prairie pothole region

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ABSTRACT

In the Prairie Pothole Region (PPR) of Canada, wetlands once utilized for agricultural purposes are restored through the placement of a ditch plug to return them to their pre-existing hydrological state. The overall objective of this research was to assess differences in riparian soil microbial community structure between reference wetlands, those which had never been utilized for agricultural purposes, and restored wetlands, of varying times since restoration. Soil samples (0–6 cm) were taken from 15 reference and 28 restored wetlands. The soil microbial community was characterized using phospholipid fatty acid (PLFA) analysis. Data were analyzed using non-metric multidimensional scaling, multivariate regression trees (MRT) and indicator species analysis. The microbial community of younger restored soils (1–3 and 4–6 yrs) differed significantly from the reference soils, with reference soils having higher microbial biomass, evenness, and diversity. Richness showed an increasing trend with time since restoration. Results from the MRT underlined the importance of climatic factors, specifically precipitation – potential evapotranspiration (P-PE) in explaining the variation found in the microbial community. More specifically, drier sites had strong indicator species values associated with PLFAs of actinomycetal origin and fungal origin. Within the wetter sites, it was found that the older restored sites (7–11 yrs) and reference sites had strong indicator species values associated with PLFAs of Gram negative and fungal origin. The similarities in microbial community composition and biomass of the older restored sites (7–11 yrs) and the reference sites indicate that this component of the wetland ecosystems begins to recover within this time period.

1. Introduction

Judgment of “success” in restoration projects is often limited to visual cues of aboveground indicators such as wildlife use and plant diversity and coverage (Mitsch and Wilson, 1996; Mummy et al., 2002). Since soils respond dynamically to stresses and disturbances, a closer look at belowground characteristics not only offers a more complete understanding of ecosystem dynamics, but also better determines the return of elements of functionality in order to gauge successful restoration efforts than more general plant and animal surveys (Harris, 2003). The soil microbial community is inherent in determining the biogeochemical cycles and organic matter turnover in soils (Zelles, 1999). It integrates both the physical and chemical aspects of the soil environment and is sensitive to anthropogenic activity, making it a suitable indicator of overall soil quality (Peacock et al., 2001a). The soil microbial community has been used in relation to various restoration projects to determine the state of the restored ecosystems when compared to “target” ecosystems (Harris, 2009). Nearby undisturbed sites are often chosen as “target” ecosystems as they may be considered biologically stable and representative of the pre-disturbance conditions (Mummy et al., 2002).

One method to examine the structural composition of the soil microbial community is phospholipid fatty acid (PLFA) analysis (Vestal and White, 1989). Because phospholipids are rapidly degraded following cell death, PLFA analysis is a good index of the living community (Bardgett et al., 1999). The measurement of PLFAs has been utilized to examine the microbial community in agricultural soils (Bossio et al., 1998; Zelles et al., 1992) and in restored prairie soils (McKinley et al., 2005). McKinley et al. (2005) found that with time the soil characteristics and the PLFA profiles of restored prairie soils became increasingly similar to that of the native soils. Hence, PLFA profiling may be considered a sensitive method for determining changes in the overall microbial community during land restoration. However, to date, very few studies have examined the changes in microbial community structure of

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restored wetlands (D'Angelo et al., 2005) and specifically the changes that occur within riparian, or hydric, soils of these restored wetlands.

The Prairie Pothole Region (PPR) extends across north-central North America and covers approximately 715,000 km^2 (Euliss et al., 1999) with over 67% of the PPR occurring in Canada. The Canadian portion of the PPR contains 80% of the agricultural land in Canada along with more than 4.5 million hectares (11 million acres) of wetlands (DUC, 2003). Land use pressure has resulted in the drainage of these wetlands, with estimates of 71–75% of wetlands being converted to agriculture throughout the Canadian PPR (Anonymous, 1986; Patterson, 1999). The loss of wetlands was first recognized due to the loss of habitat for nesting waterfowl populations and subsequent action was taken to achieve continental waterfowl population goals, primarily through the recovery of habitat. Ducks Unlimited Canada (DUC) has been the primary organization responsible for the majority of wetland restorations to date, with over 900 wetlands (or 1700 ha) in Alberta, Saskatchewan and Manitoba being restored between 1989 and 1997. Restoration designs are dependent on the modifications necessary to create the desired hydrology including excavating the basin, building dykes, removing tiles or plugging ditches (Galatowitsch and van der Valk, 1994). All of the wetlands included in the present study were restored by installing a ditch plug. In this study restoration is further defined as the reestablishment of a wetland through cessation of artificial drainage.

The overall objective of this study was to examine the microbial community structure in riparian soils of restored wetlands. Furthermore, we determined the return of key characteristics linked to microbial activity in these restored soils of varying ages since restoration (3–11 yrs) by making comparisons with undisturbed, or reference, wetlands. Since native prairie soil is renowned for its fertility, high soil organic matter content and high microbial biomass (McKinley et al., 2005), we expected that reference soils would have higher microbial biomass and diversity than restored soils. We also hypothesized that with increasing time since restoration the microbial community would become more similar to the reference soil microbial communities. Another objective of this study was to investigate the relationship of the microbial community structure to carbon characteristics (total carbon content, C/N and δ^{13}C) of the soils. Specifically, we were interested in separating distinct soil carbon pools to explore their respective relations to microbial communities.

2. Materials and methods

2.1. Study sites and sampling protocol

The Canadian PPR includes approximately 480,000 km^2 and runs through southeastern Alberta, southern Saskatchewan and southwestern Manitoba (Johnson et al., 1989). The parent material of the region consists of glacio-lacustrine sediments (Kantrud et al., 1989) and the topography of this area is mostly flat to gently rolling. Warm summers, with air temperatures throughout the region being similar with a July mean of approximately 18 °C (Greenwood et al., 1995), and cold winters, where air temperatures can drop below −60 °C (Euliss et al., 1999) characterize the PPR. Snow covers the ground 30–50% of the time and precipitation varies from 400 to 600 mm per year. The region typically experiences a negative water balance, with evaporation exceeding precipitation (Richardson et al., 2001). The PPR encompasses two physiographic zones: the aspen parkland, and prairie or grassland region (Greenwood et al., 1995). The aspen parkland is transitional between the boreal forest and prairie, where the natural vegetation is grassland interspersed with aspen and oak bluffs. The prairie or grassland region is characterized by dry mixed grassland and tallgrass prairie. The dominant soils are Udic, Typic, and Aridic Borolls (Soil Survey Staff, 2006), or Black, Dark Brown and Brown Chernozems according to the Canadian classification (Agriculture Canada, 1988a–c; Soil Classification Working Group, 1998).

A total of 43 wetlands were sampled throughout the PPR of Alberta, Saskatchewan and Manitoba in August of 2005. These wetlands were chosen from an inventory of over 898 restored wetlands based on a series of criteria, including site accessibility, age class, and presence of a reference wetland within a quarter section of the restored wetlands. The wetlands in this study were classified as either Class III or Class IV, where the ponding duration under normal conditions is 1–3 to 5 months respectively (Stewart and Kantrud, 1971). Also included in this study was one wetland in Southern Saskatchewan classified as a Class II or temporary pond. In total 15 reference wetlands and 28 restored wetlands were sampled. The classification of soil zone for each site was determined by referencing the Canada Soil Inventory Maps for each province (Agriculture Canada, 1988a–c), with gleyed variants of these soils dominating the riparian zones. Twenty-four of these wetlands were located in the aspen parkland ecoregion and Black Chernozem soil zone of Alberta (Table 1). Sixteen wetlands were located in Saskatchewan and included wetlands in the Black, Dark Gray and Brown Chernozem soil zones as well as in the aspen parkland ecoregion and prairie ecoregion. Finally, 3 wetlands were located in the Aspen Parkland ecoregion and Black Chernozem soil zone of Manitoba. The restored wetlands ranged in age from 3 to 11 years since restoration (Table 1) and all were restored with the placement of a ditch plug. After installation of the ditch plug, reseeding with a dense nesting cover mix (DNC), which is generally comprised of different species of wheatgrasses (*Agropyron* spp.), meadow brome (*Bromus biebersteinii*) and other species of native prairie grasses, occurred to provide erosion protection but no fertilization took place.

Each study site included one reference wetland, which based on air photo interpretation and discussion with land owners, had not been purposely drained for agricultural purposes, and one or more restored wetlands of the same restoration age. Within a site, wetlands varied in size but were all less than 10 ha in area (Fig. 1) and all wetlands were located on the same quarter section of land (160 acres or 2.59 km^2). A transect had been delineated at each wetland for the detailed greenhouse gas study that occurred at the sites from 2003 to 2005 (Pennock, personal communication, 2007). Soil samples for microbial community (PLFA) determination were collected using an Oakfield corer (diameter = 3.18 cm) and were taken within 1 m of the original transect position to the left or right depending on trampling or disturbance. Fresh litter and live plant material was removed from the area where the core would be taken. Four sampling points were selected within the riparian area through identification of hydrophytic vegetation and position in relation to the basin (Fig. 1). Four cores, approximately 30 cm apart, were taken at each sampling point and composited in sterile bags. Nitrile gloves were worn by samplers and equipment was wiped with 95% ethanol solution between positions to prevent contamination of the sample and cross-contamination between samples. Cores were taken to a depth of 6 cm as this is where we would expect to see most changes to soil characteristics over the short term (Karlen et al., 1999; McKinley, 2001; McKinley et al., 2005). Soil samples for pH and texture determination were collected using a bulk density corer (diameter = 7.5 cm) and followed the same procedure as outlined above. All samples were placed in a cooler and kept on ice until they were transported to the laboratory within a 12 h period. Samples for PLFA analysis were stored at −86 °C until analyzed.

Table 1
Soil classification, climatic variables and time since restoration for the study sites.

Province	Site Name	Latitude, Longitude (°N, °W)	Canadian soil Classification[a]	USDA Classification[b]	Time Since Restoration (yr)	Age Class (yrs)	P-PE[c] (mm)	EGDD[d]
Alberta	Ambler	53.09, −113.20	Black Chernozem	Udic Borolls	9	7–11	−205	1294
	Boyden	52.07, −113.18	Black Chernozem	Udic Borolls	11	7–11	−265	1242
	Ferleyko	53.54, −112.27	Black Chernozem	Udic Borolls	3	1–3	−233	1264
	Kemo	52.07, −113.18	Black Chernozem	Udic Borolls	5	4–6	−265	1242
	Loveseth	53.16, −111.50	Black Chernozem	Udic Borolls	4	4–6	−243	1270
	Maruschak	53.18, −113.12	Black Chernozem	Udic Borolls	11	7–11	−205	1294
	Mittlestadt	53.18, −113.13	Black Chernozem	Udic Borolls	11	7–11	−205	1294
	Rauser	53.11, −111.46	Black Chernozem	Udic Borolls	10	7–11	−243	1270
Saskatchewan	Adams	51.02, −101.87	Black Chernozem	Udic Borolls	7	7–11	−290	1337
	Crone	52.26, −104.71	Black Chernozem	Udic Borolls	5	4–6	−290	1352
	Hartt	53.08, −104.54	Dark Gray Chernozem	Boralfic Borolls	8	7–11	−306	1318
	Peters	50.62, −106.93	Brown Chernozem	Aridic Borolls	8	7–11	−401	1483
	Sprig	50.46, −106.33	Brown Chernozem	Aridic Borolls	8	7–11	−400	1481
	Tataryn	51.21, −103.12	Black Chernozem	Udic Borolls	3	1–3	−265	1307
Manitoba	Graham	50.21, −99.74	Black Chernozem	Udic Borolls	3	1–3	−252	1341

[a] Agriculture Canada (1988a–c).
[b] Soil Survey Staff (2006).
[c] Precipitation − potential evapotranspiration.
[d] Effective growing degree days.

2.2. Laboratory analyses

Air-dried, sieved (<2 mm) soil samples from the four positions along each transect were composited prior to analysis and pH was determined on composited samples using a 1:2 soil: 0.01 M $CaCl_2$ ratio and a settling time of 30 min (Kalra and Maynard, 1991; Peech, 1965) on a Fisher Scientific Accumet pH meter. Loss on ignition was used to estimate soil organic matter content following the procedure outlined by Ball (1964) and Kalra and Maynard (1991) using samples from reference wetlands. Based on the results from the loss on ignition method, pre-treatment of all samples to remove organic matter was carried out on soils when organic matter was determined to be greater than 5% wt (Gee and Or, 2002). Texture of the samples was then determined using the hydrometer method as outlined in Gee and Or (2002) and textural class was determined

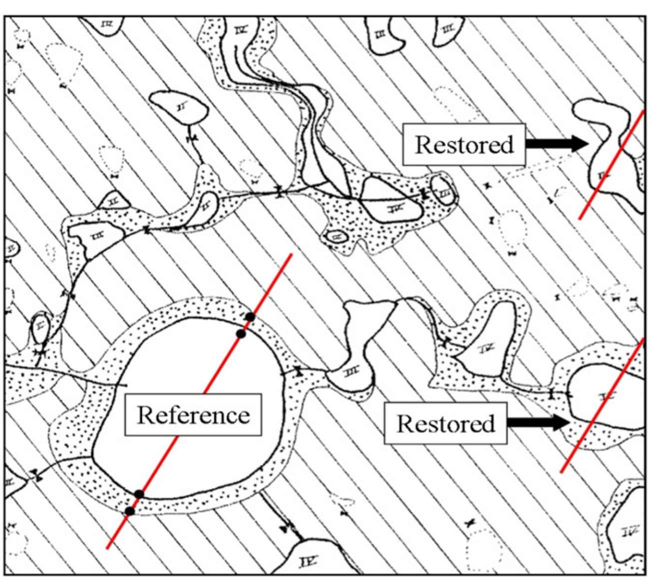

Fig. 1. Detailed map of a site, showing the reference wetland and two restored wetlands of the same age, the transects established across each wetland and the possible sampling points (as shown at the reference wetland). All wetlands at a site occur on one quarter section of land (160 acres or 2.59 km^2), with the average size of the basins generally < 2 ha.

using the Canadian System of Soil Classification (Soil Classification Working Group, 1998).

Air-dried (<2 mm) soil samples were fractionated using a combination of density (ρ), particle-size separation (μm), and acid hydrolysis (Baldock et al., 1992; Oades et al., 1987; Ryan et al., 1990) to yield the following fractions: the light fraction ($\rho < 1$ g cm^{-3} and >53 μm), the Sand fraction ($\rho > 1$ g cm^{-3} and >53 μm), the Silt + Clay fraction (<53 μm), and the acid-unhydrolyzable residue (AUR) fraction, which is the residue following digestion with H_2SO_4. Total C and N analyses were completed on all soil fractions by dry combustion using a Costech ECS 4010 CHNS-O Elemental Combustion System (Costech Analytical Technologies Inc., Valencia, CA). Values for C isotopic composition ($\delta^{13}C$), again on all soil fractions, were obtained on a Costech ECS 4010 CHNS-O Elemental Combustion System coupled to a Finnigan Deltaplus Advantage™ Isotope Ratio Mass Spectrometer (ThermoFinnigan, Bremen Germany). Results were expressed using the δ-notation, the ‰ variation from the standard Pee Dee Belemnite (PDB) reference material, and the isotopic composition calculated from:

$$\delta^{13}C = \left[\left(R_{\text{sample}}/R_{\text{standard}}\right) - 1\right]1000$$

where R_{sample} and R_{standard} are the ratios of $^{13}C/^{12}C$ in the sample and standard, respectively.

Samples for PLFA analysis were freeze-dried and then analyzed according to the procedure outlined in Hannam et al. (2006). Using a modified Bligh and Dyer (1959) extraction, lipids were extracted from 300 mg aliquots of freeze-dried mineral soil. A pre-packed silicic acid column (Agilent Technologies, Wilmington, NE), conditioned with 5 ml of chloroform, was used to extract polar lipids from neutral and glycolipids. Neutral fatty acids were eluted from the column by addition of chloroform and glycolipids were eluted with the addition of acetone. The polar lipids, which include the phospholipid fraction, were eluted from columns using methanol, which was then evaporated under nitrogen. Phospholipids were then subjected to mild alkaline methanolysis to form fatty acid methyl esters (FAMEs). Using an Agilent 6890 Series capillary gas chromatogaph (Agilent Technologies, Wilmington, NE) equipped with a 25 m Ultra 2 (5%-phenyl)-methylpolysiloxne column the FAMEs were separated and quantified. Hydrogen was used as the carrier gas. Peaks were identified based on bacterial fatty acid standards and MIDI peak identification software (MIDI, Inc., Newark, DE).

Fatty acids were designated as X:YωZ, with X indicating the number of carbon atoms, Y indicating the number of double bonds, and Z indicating the position of the first double bond from the aliphatic (ω) end of the molecule. The suffixes 'c' and 't' indicate cis and trans geometry. The prefixes 'i', 'a', and 'me' refer to iso, anteiso and mid-chain methyl branching, and 'cy' refers to cyclopropyl rings.

2.3. Statistical analyses

Based on a preliminary analysis of the data performed to test for differences among all restoration ages, three age groupings were selected to reflect restoration stages: 1–3, 4–6, and 7–11 years. Data corresponding to similar positions on the transect in relation to the basin were combined to yield two separate transect positions per wetland. Differences between transect positions as well as among age groups in the microbial community, as defined by the microbial indices (Table 2) were analyzed using a two-way ANOVA with PROC GLM (version 9.1, SAS Institute Inc.). Position will not be discussed in the remainder of this manuscript as there were no significant effects of position and no significant interactions between position and age groups. All indices were found to meet the assumptions of normality. Post-hoc comparisons were performed using the SNK test and determined to be significant at $p \leq 0.05$. Microbial community structure (PLFA) patterns in reference and restored wetland soils were further analyzed through a non-metric multidimensional scaling (NMS) ordination technique, followed by multi-response permutation procedures (MRPP), using the PC-ORD software (version 4, MjM Software Design, Gleneden Beach, OR). Ordination allows items to be graphically organized as to summarize complex relationships and then to extract dominant patterns from many possible outcomes (McCune and Grace, 2002). In particular, NMS runs numerous iterations from which the best possible way to represent the data is reduced to either two or three dimensions, with the distance between the data points indicative of the similarity between those points. NMS does not require normal distribution of data nor does it assume a linear relationship between variables. The Sorensen (Bray-Curtis) distance was used for these analyses. All parameters were tested for significance in the NMS analysis using a multi-response permutation procedure (MRPP). MRPP is a non-parametric procedure used to determine if there are differences between 2 or more groups (McCune and Grace, 2002). Significance was determined from the output of three values: the p value, which indicates overall significance of the comparisons; the test statistic (T) which specifies the separation between groups, with a more negative T value indicative of stronger separation; and the agreement statistic (A), which indicates the within group homogeneity versus the random expectation, with a higher A value signifying higher homogeneity.

Table 2
Microbial indices.

Indices	PLFA marker
Total Biomass	Sum of all fatty acids, in nmol/g soil (DW)
Diversity	Shannon Index (H′), $\Sigma(p_i(\ln p_i))$
Richness	Total number of non-zero PLFAs with total carbon length <20
Evenness	Measure of the variability in abundance of different PLFAs within a sample, Diversity/ln(Richness)
% Gram (+)[a, b, c]	a11:0, i14:0, i15:0, a15:0, i16:0, a17:0, i17:0/Total Biomass
% Gram (−)[a, b, d, e]	14:1ω5c, 16:1ω5c, 16:1ω9c, 18:1ω5c, 18:1ω7c, 18:1 9c/Total Biomass
% Fungi[f, g, h]	18:2ω6,9c/Total Biomass
% Actinomycetes[i]	10Me18:0/Total Biomass
Sulfate reducing bacteria[j, k]	10 Me16:0/Total Biomass
Sulfate reducing bacteria[l]	17:1ω8/Total Biomass
Anaerobic bacteria[m]	cy19/Total Biomass
Fungal/Bacterial biomass[i, n, o]	18:2ω6,9c/i15:0, a15:0, 15:0, i16:0, 16:1ω9, 16:1ω5, i17:0, a17:0, cy17:0, 17:0, 18:1ω7, and cy19:0
Gram (+)/Gram (−)[p]	a11:0, i14:0, i15:0, a15:0, i16:0, a17:0, i17:0/14:1ω5c, 16:1ω5c, 16:1ω9c, 18:1ω5c, 18:1ω7c, 18:1ω9c
iso to anteiso[p]	i15:0, i17:0/a15:0, a17:0

a Mentzer et al. (2006).
b O'Leary and Wilkinson (1988).
c D'Angelo et al. (2005).
d Sundh et al. (1997).
e Wilkinson (1988).
f Lechevalier and Lechevalier (1988).
g Frostegard and Baath (1996).
h Zelles (1999).
i Frostegard et al. (1993).
j Dowling et al. (1986).
k Parkes and Taylor (1983).
l Bossio and Scow (1998).
m Guckert et al. (1985).
n Tunlid et al. (1989).
o DeGrood et al. (2005).
p McKinley et al. (2005).

The first matrix for analysis of the microbial community data contained all PLFA measured and was expressed on a mol% basis, relativized by row, and transformed using the arcsine square-root function. The second matrix contained chosen site and soil descriptive variables in order to map these vectors over the first matrix: time since restored (yr), age group, position on transect, soil zone, textural class, wetland class, pH, and climatic data. Climatic data consisted of precipitation − potential evapotranspiration (P-PE) and effective growing degree days (EGDD) values derived from 10 km × 10 km data grids (Canadian Climate Impact Scenarios Project, 2005). The second matrix for the microbial community data also contained parameters derived from PLFA analysis, including all calculated microbial indices (Table 2). It should be noted that the use of diversity, evenness and richness indices needs to be interpreted with caution as the resolution of PLFA profiling is lower than the actual soil microbial community. In addition, carbon characteristics such as organic carbon (OC) concentrations, C/N ratios, and $\delta^{13}C$ values for the bulk soil, light fraction (LF), sand-sized, silt-clay sized and acid-unhydrolyzable (AUR) fractions were included.

Further analysis of the data included the use of multivariate regression trees (MRT) using the R package (version 2.8.0, The R Foundation for Statistical Computing) and the mvpart library (Therneau and Atkinson, 2005). Trees were constructed to compare the microbial community structure (PLFA) to categorical environmental variables (De'ath, 2002). In brief, MRT is a non-parametric method which can handle non-linear relationships and complex ecological datasets; it divides data into groups using least square splitting criteria based on the environmental variables so that the dissimilarity between groups is maximized and dissimilarity within groups is minimized. Each split in the MRT is represented graphically as a branch in the tree, and the length of each branch represents the variation in the data explained by that split.

The categorical environmental variables included in the MRT analysis were age groups (1–3 yrs, 4–6 yrs, 7-11 yrs and reference wetlands), position on transect, P-PE, textural class and wetland class. The species data used were the same mol% PLFA data used in the ordination as described above. The groupings of PLFAs resulting from the MRT analysis were then tested in ordination space using MRPP, as described above. Indicator species analysis was utilized to determine which species characterized the soil microbial community of grouping variables based on MRT clusters (McCune and Grace, 2002). Finally, these indicator values were tested for significance using a randomized Monte Carlo technique.

3. Results

Analysis of the individual microbial indices (Table 2) by age groups showed significant differences between the reference sites, and the 1–3 yr and 4–6 yr restored sites. Specifically, the reference sites had higher values for PLFA biomass (calculated based on the sum of 96 PLFAs), evenness and diversity (Fig. 2). No significant differences were found between the older restored sites (7–11 yrs) and the reference sites for any of the indices calculated. Although no significant difference was found in PLFA richness between age groups the graph shows an increasing trend with age. A significant difference in the percent of actinomycetes was found between the reference sites and the 4–6 yr age grouping, with the 4–6 yr age grouping having a significantly higher percentage of actinomycetes than the reference sites ($p = 0.009$). No other significant differences among age groups were found for any of the other microbial indices calculated, including those indicative of stress and sulfate reducing or anaerobic bacteria.

MRTs with endings of five groups were consistently produced with multiple cross validations and explained 54% of the variation in the PLFA data (Fig. 3). After 1000 cross validations, trees with five branches occurred 294 times. The first branching of the MRT was caused by the climatic factor precipitation − potential evapotranspiration (P-PE), which explained 23% of the variation. Groups 1 and 2 consisted of the "wet" sites, with the P-PE ranging from −205 mm to −252 mm. These groups in the "wet" sites were split further by age groups, which explained 29% of the variation and divided between sites 1–3 yrs and 4–6 yrs restored, and 7–11 yrs restored and reference sites. The "dry" sites consisted of groups 3, 4 and 5 with the P-PE ranging from −265 mm to −401 mm. The groupings in the "dry" sites were split further by textural class, which explained 26% of the variation and separated the clay loam (average of 33% sand and 31% clay) and silt loam (average of 32% sand and 12% clay) soils from the loamy textured soils (average of 42% sand and 21% clay). Finally, the last split of the loamy soils was due to age groups, which explained 58% of the variation and divided the 1–3 and 4–6 yr restored sites from the 7–11 yr and reference sites.

The NMS ordination of microbial community structure produced a three-dimensional solution with a final stress of 10.3 after 135 iterations (Fig. 4). Axis 2 (18%) and 3 (35%) represented the

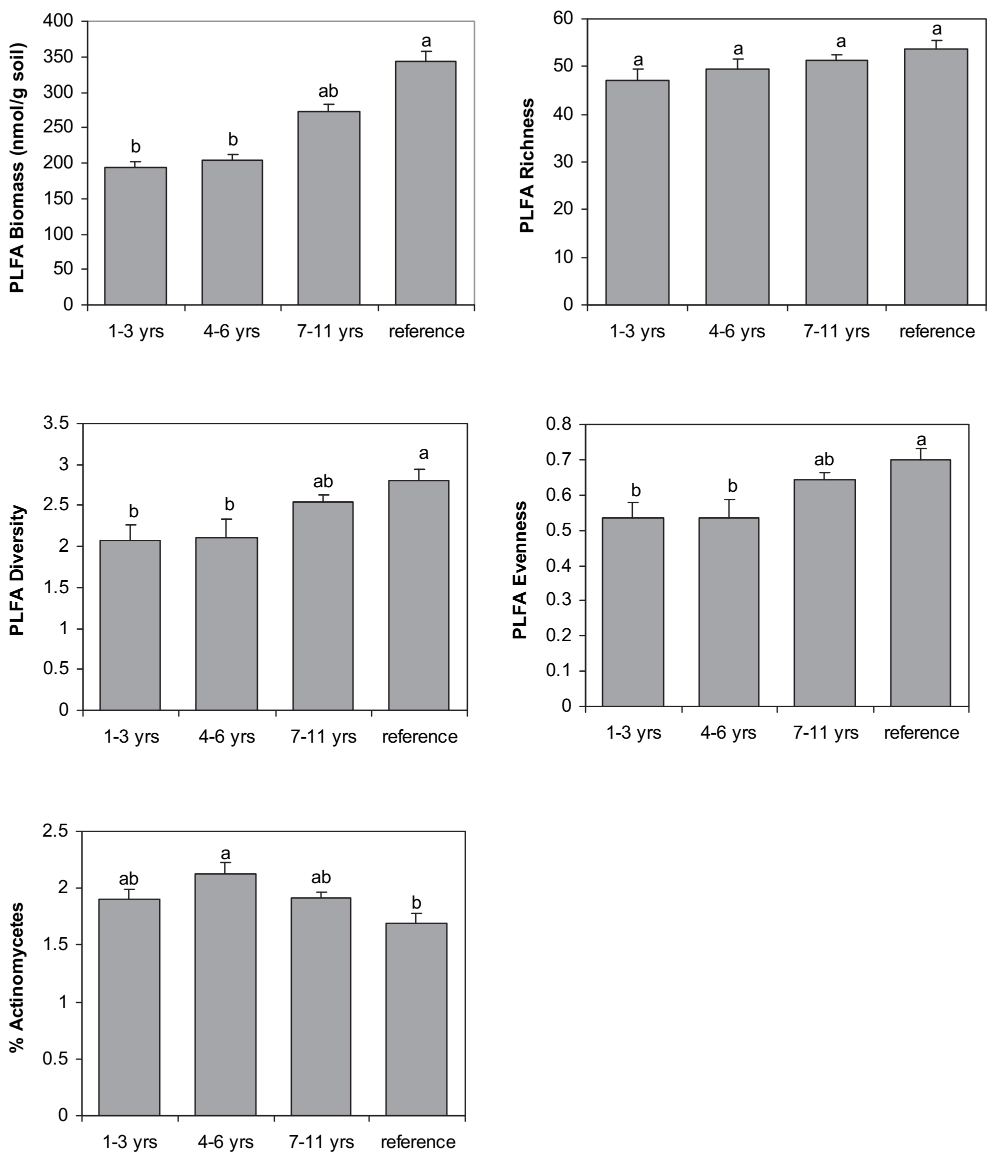

Fig. 2. Graphs of PLFA indices (mean and standard error bars with $n = 8$ for 1–3 yrs, $n = 11$ for 4–6 yrs, $n = 31$ for 7–11 yrs, and $n = 25$ for Reference sites) comparing the restored wetland soils of varying times since restoration and reference wetland soils. Bars with the same letter are not significantly different based on SNK test ($P < 0.05$).

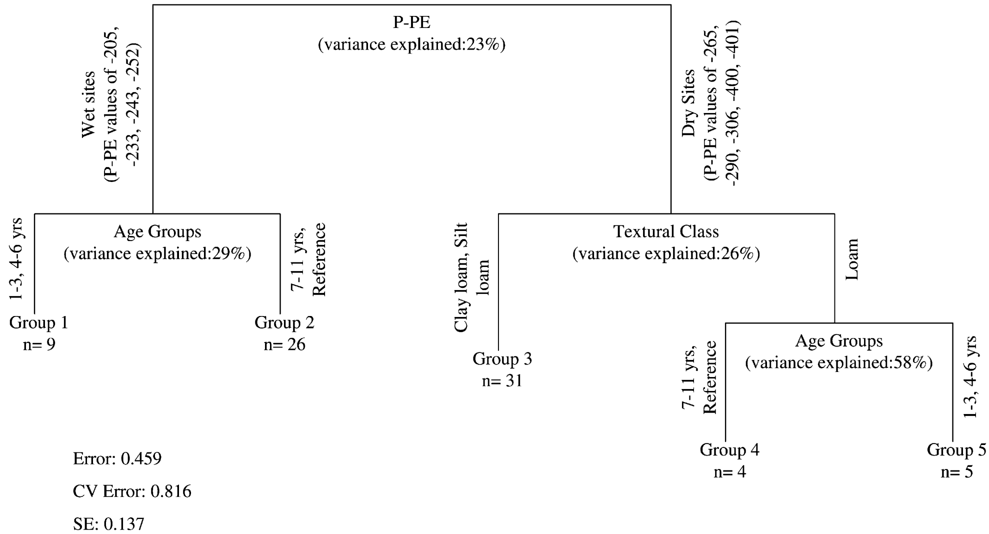

Fig. 3. Multivariate regression tree of mol% PLFA data from riparian soils of restored and reference wetlands. Groups are combinations of age groups (1–3 yrs, 4–6 yrs, 7–11 yrs, reference), climate (P-PE/Precipitation- potential evapotranspiration) and textural class.

strongest correlation structure in the data and provided the most interpretable ordination. The MRPP analysis for time since restoration (Table 3) showed both strong separation between groups and strong within group homogeneity ($T = -9.67$, $A = 0.11$, $p = < 10^{-8}$). Significant differences in microbial community structure between sites of varying time since restoration were found (Table 3), with the reference and older sites (9, 10, and 11 yrs) grouping closer in ordination space, and farther away from the younger sites (3 yrs). This particular ordination graph was not included here as a clearer interpretation of the data set occurred after inclusion of groups derived from the MRT (Fig. 4). When the MRPP was re-run on the groupings from the MRT, larger significant differences were found (Table 3). In particular, the largest difference based on the MRPP was associated with the climatic variable corresponding to precipitation − potential evapotranspiration (P-PE), which had the lowest T value (−17.98), highest A value (0.14) and most highly significant p-value (1.0×10^{-6}). A significant difference was also found within the soil microbial community of the "wet" sites (P-PE ranging from −205 mm to −252 mm), as shown on Fig. 4, between Group 1, the younger restored sites (1–3 and 4–6 yrs) and Group 2, the older restored (7–11 yrs) and reference sites. In addition, a significant difference was found based on textural class with Group 3 (clay loam and silt loam textures), separating from the loamy textured Groups 4 and 5.

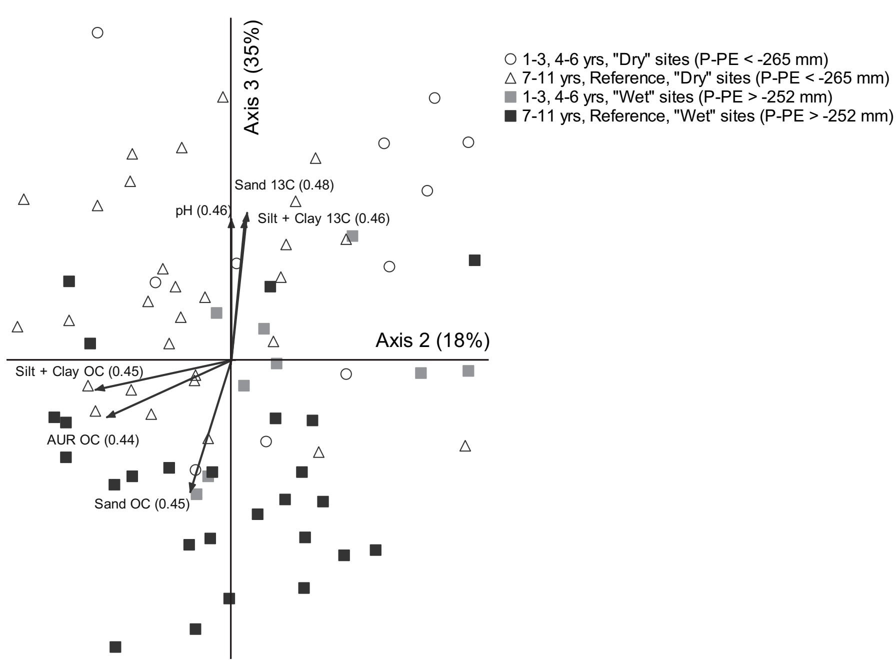

Fig. 4. NMS ordination of wetland riparian soil PLFAs delineated by clusters based on results from the multivariate regression tree (Fig. 3). Ordination points labeled as ○ and △ represent the "dry" sites (Groups 3,4,5) and points labeled as ■ and ■ represent the "wet" sites (Groups 1 and 2). Dry sites are those with P-PE (Precipitation − potential evapotranspiration) < −265 mm. Carbon vectors include $\delta^{13}C$ values and OC (organic carbon) values of various fractions (Sand, Silt + Clay and AUR, or acid-unhydrolyzable residue), with r^2 values in parentheses.

Table 3
Multi-response permutation procedure (MRPP) results for microbial community structure based on PLFA analysis, including groupings based on MRT results reported in Fig. 3.

Comparison		T[a]	A[b]	P[c]
Time Since Restored		−9.67	0.11	$<10^{-8}$
MRT group				
P-PE[d]	1,2 vs. 3,4,5	−17.98	0.14	$<10^{-6}$
Age Group	1 vs. 2	−8.61	0.13	$7.5 * 10^{-5}$
Texture	3 vs. 4,5	−5.73	0.086	$1.4 * 10^{-3}$

[a] Test statistic (T) which specifies the separation between groups.
[b] Agreement statistic (A), which indicates the within group homogeneity versus the random expectation.
[c] Indicates overall significance of the comparisons.
[d] Precipitation − potential evapotranspiration.

The relationships of the microbial community structure to environmental conditions including pH and soil carbon characteristics was interpreted through a correlation analysis, where vectors were considered significant at $r^2 = 0.4$ (Fig. 4). These vectors included the organic carbon (OC) concentrations of the sand fraction, which was associated with the "wet" sites (P-PE > -252 mm), while the $\delta^{13}C$ (13C) of the sand and silt+ clay fractions, and pH were associated with the "dry" sites (P-PE < -265 mm).

Strong indicator species association values were found for all of the major branches indicated in the MRT (Table 4). The primary separation in the MRT, P-PE, was found to have two biomarkers with strong indicator species association values for the "dry" sites, one of actinomycetal origin (10ME18:0), and one of fungal origin (18:2ω6:9c). Within the "wet" sites, an indicator value associated with Gram (-) bacteria (14:1ω5c) and an indicator value for fungi (18:2ω6:9c) was strongly associated with the older restored (7–11 yrs) and reference sites. The same two biomarkers that were strongly associated with the "dry" sites in the primary P-PE branching, one of actinomycetal origin (10ME18:0), and one of fungal origin (18:2ω6:9c), were also found to be strongly associated with the clay loam and silt loam branch of the MRT.

4. Discussion

Distinct differences in microbial community composition between reference and restored wetlands of varying ages were found in this study. Results from the PLFA analysis show that the microbial community in restored wetland riparian soils approaches that of reference wetland riparian soils after seven years in terms of PLFA biomass, diversity, evenness, and richness (Fig. 2). Total PLFA biomass has previously been found to correlate positively with time since restoration in a study comparing a virgin prairie grassland with a recently (7 yr) and older (24 yr) restored agricultural site (McKinley et al., 2005). Total PLFA biomass also was reported to be higher in reference soils and those that experienced lower levels of anthropogenic disturbance (Peacock et al., 2001a). Similarly, microbial diversity and evenness were also found to be higher in an older created wetland (8 yr) when compared with 1 to 5 year old created wetlands (Ahn and Peralta, 2009).

Table 4
PLFA indicator species associated with the multivariate regression tree (MRT) groups as reported in Fig. 3.

PLFA	Origin	Mean[a]	Indicator value MRT Group		Monte Carlo ($p < 0.05$)
P-PE[b]					
			1,2	3,4,5	
10ME18:0	Actinomycete	52.2(1.77)	38	62	0.0002
18:2ω6:9c	Fungi	52.8(2.18)	38	62	0.001
Age groups					
			1	2	
14:1 ω5c	Gram (−)	49.3(4.97)	13	73	0.0002
18:2ω6:9c	Fungi	54.3(3.15)	29	71	0.0002
Texture					
			3	4,5	
10ME18:0	Actinomycete	52.9(2.31)	63	37	0.0006
18:2ω6:9c	Fungi	53.8(2.84)	65	35	0.0006

[a] Means and standard deviations in parentheses are based on Monte Carlo test of observed indicator values for each species based on 1000 randomizations (McCune and Grace, 2002).
[b] Precipitation − potential evapotranspiration.

Results from the MRT analysis indicated that the primary influence over the soil microbial community composition was the climatic variable, P-PE (Fig. 3). Specifically, sites that had more negative P-PE values were significantly different from the sites with less negative P-PE values, indicating that significant differences existed in the hydrologic conditions at the sites. Moisture is a well known determinant of soil microbial community composition as it regulates the rates of aerobic or anaerobic processes in the soil and the resulting community structure (Gutknecht et al., 2006). Two specific biomarkers, one of actinomycetal origin (10ME18:0) and one of fungal origin (18:2ω6:9c) were found to be indicator species for the "dry" sites, or those sites with more negative P-PE values. Flooding has been shown to decrease the abundance of fungi in an agricultural field (Bossio and Scow, 1998) and in a freshwater wet prairie ecosystem (Mentzer et al., 2006). The finding in the present study, that fungi are more abundant in drier ecosystems, mirrors findings in the studies mentioned above (Bossio and Scow, 1998; Mentzer et al., 2006). Fungi are also known to withstand higher levels of osmotic stress than bacteria in tallgrass prairie soils (McKinley et al., 2005), thus making them more likely to prevail in dry environments. Furthermore, a significantly higher abundance of actinomycetes was found in the 4–6 year restored riparian soils, compared with the reference riparian soils (Fig. 2). Peacock et al. (2001a) similarly found that heavily trafficked areas contained more PLFAs associated with actinomycetes and attributed this to their ability to grow conidia and thus survive harsh soil conditions. The lower P-PE values found at the sites with actinomycete biomarkers as indicator species (Fig. 3) are indeed consistent with a harsher soil environment. Further, we would expect that the harshest soil environmental conditions, increased heat and desiccation, would be demonstrated at the younger (1–3 yrs and 4–6 yrs) restored sites as hydrologic conditions, plant community and soil chemical properties may not have yet recovered from the agricultural disturbance.

The branch in the MRT that included all the "dry" sites was further divided into two leaves separated based on texture. It was found that clay loam and silt loam soils had significantly different microbial communities than the loamy soils (Fig. 3). Further investigation using indicator species analysis (ISA) found that an actinomycetal biomarker (10ME18:0) and a fungal biomarker (18:2ω6:9c) were strong indicators for the clay loam and silt loam textured soils. Although the groups 3, 4 and 5 in the MRT are all considered "dry", the increased desiccation due to the higher percentage of sand in the loam soil has seemingly resulted in the microbial community reaching an upper limit of tolerance of desiccation and heat, as both fungi and actinomycete are able to withstand these harsh conditions better than other microbes.

Within the "wet" sites, the microbial community was split further to show significant differences between the younger restored sites (1–3 and 4–6 yrs), and the older restored (7–11 yrs) and reference sites (Fig. 3). This supports our hypothesis that with increasing time since restoration the microbial community became more similar to the reference microbial community. Microbial communities have been reported to be significantly different between early and late successional wetlands (D'Angelo et al., 2005), between virgin prairie, restored prairie and agricultural sites (McKinley, 2001), and between agricultural sites, recent and long-term restorations and virgin prairie sites (McKinley et al., 2005). Two biomarkers, a gram negative bacterial biomarker (14:1ω5c) and a fungal biomarker (18:2ω6:9c), were found to be strong indicator species for the older restored (7–11 yrs) and reference sites (Table 4). In contrast to our findings, D'Angelo et al. (2005) found that the same indicators of gram negative bacteria and fungi, both representative of aerobic microbial communities, were both more abundant in early successional wetlands (<10 years since mitigation) and attributed this to oxygen availability being higher at the early sites compared to the late successional wetlands, which experienced higher instances of anaerobiosis. This suggests that differences observed in our study between the younger restored sites, and the older restored and reference sites (Fig. 3) were not primarily influenced by the hydrologic conditions at the sites. Instead, the longer period without disturbance, found in the 7–11 yr sites, and the absence of disturbance found in the reference sites may be the dominating influence. Lower levels of Gram negative bacterial populations have been associated with soils more recently subjected to disturbance (Peacock et al., 2001a) and disturbance is known to be extremely detrimental to fungal populations (Mummy et al., 2002).

The NMS ordination of the soil microbial community (Fig. 4) showed that the vectors for pH and $\delta^{13}C$ were strongly correlated with the "dry" and younger restored sites (1–3 and 4–6 yrs). In our study, differences in $\delta^{13}C$ may be attributable to differences in decomposition processes since litter inputs did not differ among sites (Card, 2010). The ^{13}C enrichment of soil organic matter during decomposition has been linked to discrimination against ^{13}C during the catabolic breakdown of organic substrates by soil microbes and/or accumulation of ^{13}C in microbial biomass and in humic substances of microbial origin (Quideau et al., 2003; Hannam et al., 2005). Hence, the higher $\delta^{13}C$ values (i.e.; more enriched in the heavier isotope) in the "dry" and younger restored sites may indicate organic material at a more advanced stage of decomposition. The more highly decomposed material and lower carbon availability at these sites hence could have impacted the structure of the microbial community, which was also lower in biomass and lower in diversity (Fig. 2). Microbial biomass has been found to decrease with increasing acidity in an agricultural setting because most bacteria have pH optima between 6.0 and 7.5 (McKinley, 2001). However, in our study, the higher microbial biomass occurs in the "wet" sites which are lower in pH (5.8) than the "dry" sites (pH = 6.6). It is likely that a combination of soil moisture, pH and carbon availability impacted the microbial community, with not just one factor being predominant. Also, it is interesting to note that actinomycetes, an indicator species for the "dry" sites, have been found to tolerate soils with higher pH values and be sensitive to soil acidity (Alexander, 2005; Frostegard et al., 1993; Pennanen, 2001).

Composition of the soil microbial community at the older restored (7–11 yrs) and reference sites was correlated with higher OC concentrations for the sand fraction (Fig. 4). It has been well documented that cultivation practices degrade soil aggregates and accelerate microbial activity therefore increasing the turnover of soil carbon (Golchin et al., 1995). Numerous studies have shown that with longer periods without disturbance, agricultural land allowed to revert back to natural vegetation as well as restored wetlands that were previously drained exhibit an increase in soil organic carbon (Bruland and Richardson, 2006; D'Angelo et al., 2005; Galatowitsch and van der Valk, 1996; Lopes de Gerenyu et al., 2008; Schnitzer et al., 2006). In agreement with the carbon vectors discussed above (Fig. 4), gram-negative bacteria have been found to increase with an accumulation of soil carbon (Peacock et al., 2001b). In our study, the gram-negative bacteria (14:1ω5c) was found to be an indicator species for the older restored and reference sites (Table 4).

5. Conclusions

In this study, the examination of the soil microbial community indicated that there were distinct differences between the younger (1–3 yrs, 4–6 yrs) restored wetland riparian soils and the older restored (7–11 yrs) and reference soils. These differences were apparent even with significant influences of climate (P-PE) and texture. Specifically, the microbial communities of the older restored soils and the reference soils were found to have similar PLFA biomass, evenness, and diversity. Similar PLFA indicator species were found in the older restored and reference sites, demonstrating that the wetland ecosystem begins to recover within this time period and overcome the effects of agricultural disturbance. On the other hand, the distinct dissimilarity between the younger restored soils and the reference soils indicates the need for further investigation into the ecological differences between the soils to determine what management changes could potentially be made to accelerate the restoration process.

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